

List of Definitions.

(1)

"The Leading Term of the Yang-Mills Free Energy"

Abstract.

- "leading term of the free energy of three-dimensional $U(N)$ lattice gauge theory" (1)
- lattice spacing (1)
- phase cell renormalization (1)
- "the four dimensional theory" (1)
- weak coupling limit (1)

1. Introduction (all page 1)

- "quantum Yang-Mills theories" $\longleftrightarrow$
"quantum gauge theories"
- Standard Model of quantum mechanics
- "Lattice gauge theories"
- "3 and 4 dimensional quantum Yang-Mills theories in the continuum"
- "Constructive quantum field theory"
- "ultraviolet stability of 3 and 4 dimensional lattice gauge theories"
- "leading term of the free energy of 3 dimensional $U(N)$ Yang-Mills theory"
- weak coupling limit
- random matrix theory
- Selberg-type integrals